L J Institute of Engineering and Technology, Ahmedabad

Theory Test (T1,T2,T3,T4) Syllabus and Marks Distribution Details_SY_SEM-IV_Batch 2024
CE/IT/CSD/AIML/AIDS/RAI/CSE/CST/CS&IT/CEA/MA&CP/CSE(AI)/ CSE(DS)/CSE(CS)

Discrete Mathematics
117013491(CE),117023491(IT),117033491(CSD),117043491(AIML),117053491(AIDS),117063491(RAI),117123491(CSE),117133491(CST),117143491(CSIT), 117153491(CEA),017163491(MA&CP),017173491(CSE_AI),017183491(CSE_DS),17193491(CSE_CS)

Subject	Evaluation Component	Test		Course Outcome	Syllabus			Marks Weightage				Total	Duration of Test
					Unit No	Unit Name	% of Syllabus	MCQ	Theory Descriptive (Mainly Programs)	Formulas and Derivation	Numerical		
Discrete Mathematics	Continuous and Comprehensive Evaluation (CCE)	T1		CO1	1	Set Theory	8%	5	0	0	20	25	1.25
					2	Functions and Counting	9%						
					3	Propositional Logic	10%						
		T1 Test Syllabus:					27%	T1 Test Marks: 25					
		T2		CO2	4	Propositional and Predicate Logic	8%	5	0	0	20	25	1.25
					5	Recurrence Relations	7%						
					6	Relations	11%						
		T2 Test Syllabus:					26%	T2 Test Marks: 25					
		T3		CO3	7	Partial Ordering	13%	5	0	0	20	25	1.25
					8	Algebraic Structures	11%						
		T3 Test Syllabus:					24%	T3 Test Marks: 25					
		Continuous and Comprehensive Evaluation (CCE) Total T1 to T3 Tests :					77%	15	0	0	60	75	
	Semester End Evaluation (SEE)	T4	T1	CO1	1,2,3	Set Theory Functions and Counting Propositional Logic	27%	3	0	0	0	3	2
			T2	CO2	4,5,6	Propositional and Predicate Logic Recurrence Relations Relations	26%	3	0	0	0	3	
			T3	CO3	7,8	Partial Ordering Algebraic Structures	24%	3	0	0	0	3	
			T4	CO4	9	Graph Theory	12%	8	0	0	33	41	
					10	Network Flows and Trees	11%						
		T4 Test Syllabus:					100%	T4 Test Marks: 50					
		Semester End Evaluation (SEE) Total T4 Test:					23%	17	0	0	33	50	

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CE/IT/CSD/AIML/AIDS/RAI/CSE/CST/CS&IT/CEA/MA&CP/CSE(AI)/ CSE(DS)/CSE(CS)

Full Stack Development -2

117013492(CE), 117023492(IT), 117033492(CSD), 117043492(AIML), 117053492(AIDS), 117063492(RAI), 117123492(CSE), 117133492(CST), 117143492(CSIT), 117153492(CEA), 017163492(MA&CP), 017173492(CSE_AI), 017183492(CSE_DS), 017193492(CSE_CS)

Subject	Evaluation Component	Test		Course Outcome	Syllabus			Marks Weightage				Total	Duration of Test
					Unit No	Unit Name	% of Syllabus	MCQ	Theory Descriptive (Mainly Programs)	Formulas and Derivation	Numerical		
Full Stack Development - 2	Continuous and Comprehensive Evaluation (CCE)	T1		CO1	1	JSON(Javascript Object Notation)	5%	10	15	0	0	25	1.25
					2	Node JS- Introduction and Core modules	10%						
					3	Node JS Modules & Server Creation	10%						
		T1 Test Syllabus:					25%	T1 Test Marks: 25					
		T2		CO2	4	Express JS Fundamentals	5%	10	15	0	0	25	1.25
					5	Express State Management and API	10%						
					6	Express – Advanced Concept	10%						
		T2 Test Syllabus:					25%	T2 Test Marks: 25					
		T3		CO3	7	React Fundamentals & Core Concepts	10%	10	15	0	0	25	1.25
					8	React Hooks & API Integration	15%						
		T3 Test Syllabus:					25%	T3 Test Marks: 25					
		Continuous and Comprehensive Evaluation (CCE) Total T1 to T3 Tests :					75%	30	45	0	0	75	
	Semester End Evaluation (SEE)	T4	T1	CO1	1,2,3	JSON(Javascript Object Notation) Node JS- Introduction and Core modules Node JS Modules & Server Creation	25%	3	0	0	0	3	2
			T2	CO2	4,5,6	Express JS Fundamentals Express State Management and API Express – Advanced Concept	25%	3	0	0	0	3	
			T3	CO3	7,8	React Fundamentals & Core Concepts React Hooks & API Integration	25%	3	0	0	0	3	
			T4	CO4	9	MongoDB- Queries and operators	10%	16	25	0	0	41	
					10	Mongoose & MERN Integration	15%						
		T4 Test Syllabus:					100%	T4 Test Marks: 50					
		Semester End Evaluation (SEE) Total T4 Test:					25%	25	25	0	0	50	

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CE/IT/CSD/AIML/AIDS/RAI/CSE/CST/CS&IT/CEA/MA&CP/CSE(AI)/ CSE(DS)/CSE(CS)

Computer Organization and Architecture
017013493(CE), 017023493(IT), 017033493(CSD), 017043493(AIML), 017053493(AIDS), 017063493(RAI), 017123493(CSE), 017133493(CST), 017143493(CSIT), 017153493(CEA), 017163493(MA&CP), 017173493(CSE_AI), 017183493(CSE_DS), 017193493(CSE_CS)

Subject	Evaluation Component	Test		Course Outcome	Syllabus			Marks Weightage				Total	Duration of Test
					Unit No	Unit Name	% of Syllabus	MCQ	Theory Descriptive (Mainly Programs)	Formulas and Derivation	Numerical		
Computer Organization and Architecture	Continuous and Comprehensive Evaluation (CCE)	T1		CO1	1	Data Representation	2%	10	5	0	10	25	1.25
					2	Register Transfer and microoperations	10%						
					3	Basic Computer Organization	12%						
		T1 Test Syllabus:					24%	T1 Test Marks: 25					
		T2		CO2	4	Basic Computer Design	8%	10	5	0	10	25	1.25
					5	Programming the basic computer	18%						
		T2 Test Syllabus:					26%	T2 Test Marks: 25					
		T3		CO3	6	Central Processing Unit	12%	10	0	0	15	25	1.25
					7	Pipeline Processing	10%						
					8	Computer Arithmetic	5%						
		T3 Test Syllabus:					27%	T3 Test Marks: 25					
		Continuous and Comprehensive Evaluation (CCE) Total T1 to T3 Tests :					77%	30	10	0	35	75	
	Semester End Evaluation (SEE)	T4	T1	CO1	1,2,3	Data Representation Register Transfer and microoperations Basic Computer Organization	24%	3	0	0	0	3	2
			T2	CO2	4,5	Basic Computer Design Programming the basic computer	26%	3	0	0	0	3	
			T3	CO3	6,7,8	Central Processing Unit Pipeline Processing Computer Arithmetic	27%	3	0	0	0	3	
			T4	CO4	9	Memory Organization	14%	25	8	0	8	41	
					10	Input-Output Organization	9%						
		T4 Test Syllabus:					100%	T4 Test Marks: 50					
		Semester End Evaluation (SEE) Total T4 Test:					23%	34	8	0	8	50	

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CE/IT/CSD/AIML/AIDS/RAI/CSE/CST/CS&IT/CEA/MA&CP/CSE(AI)/ CSE(DS)/CSE(CS)

Theory of Computation
017013494(CE), 017023494(IT), 017033494(CSD), 017043494(AIML), 017053494(AIDS), 017063494(RAI), 017123494(CSE), 017133494(CST), 017143494(CSIT), 017153494(CEA), 017163494(MA&CP), 017173494(CSE_AI), 017183494(CSE_DS), 017193494(CSE_CS)

Subject	Evaluation Component	Test		Course Outcome	Syllabus			Marks Weightage				Total	Duration of Test
					Unit No	Unit Name	% of Syllabus	MCQ	Theory Descriptive (Mainly Programs)	Formulas and Derivation	Numerical		
Theory of Computation	Continuous and Comprehensive Evaluation (CCE)	T1		CO1	1	Introduction	2%	7	0	0	18	25	1.25
					2	Regular Expressions and Languages	5%						
					3	Finite Automata 1	9%						
					4	Finite Automata 2	11%						
		T1 Test Syllabus:					27%	T1 Test Marks: 25					
		T2		CO2	5	Deterministic Finite Automata & Non - Deterministic Finite Automata	12%	7	0	0	18	25	1.25
					6	Operations on Finite Automata	11%						
		T2 Test Syllabus:					23%	T2 Test Marks: 25					
		T3		CO3	7	Context Free Grammar and Languages	17%	8	0	0	17	25	1.25
					8	Push down Automata	13%						
		T3 Test Syllabus:					30%	T3 Test Marks: 25					
		Continuous and Comprehensive Evaluation (CCE) Total T1 to T3 Tests :					80%	22	0	0	53	75	
	Semester End Evaluation (SEE)	T4	T1	CO1	1,2,3,4	Introduction Regular Expressions and Languages Finite Automata 1 Finite Automata 2	27%	3	0	0	0	3	2
			T2	CO2	5,6	Deterministic Finite Automata & Non - Deterministic Finite Automata Operations on Finite Automata	23%	3	0	0	0	3	
			T3	CO3	7,8	Context Free Grammar and Languages Push down Automata	30%	3	0	0	0	3	
			T4	CO4	9	Turing Machine	13%	13	0	0	28	41	
					10	Recursively Enumerable Languages	7%						
		T4 Test Syllabus:					100%	T4 Test Marks: 50					
		Semester End Evaluation (SEE) Total T4 Test:					20%	22	0	0	28	50	

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Fundamentals of Computer Science using Python - II
117012491(CE), 117022491(IT), 117032491(CSD), 117042491(AIML), 117052491(AIDS), 117062491(RAI), 117122491(CSE), 117132491(CST), 117142491(CSIT), 117152491(CEA), 017162491(MA&CP), 017172491(CSE_AI), 017182491(CSE_DS), 17192491(CSE_CS)

Subject	Evaluation Component	Test		Course Outcome	Syllabus			Marks Weightage				Total	Duration of Test
					Unit No	Unit Name	% of Syllabus	MCQ	Theory Descriptive (Mainly Programs)	Formulas and Derivation	Numerical		
Fundamentals of Computer Science using Python - II	Continuous and Comprehensive Evaluation (CCE)	T1		CO1	1	Data Analysis with Pandas & EDA	13%	10	15	0	0	25	1 (Offline) 1.25 (Online)
					2	Data Visualization with Python	8%						
					3	Introduction to Machine Learning with Python	4%						
		T1 Test Syllabus:					**25%**	**T1 Test Marks: 25**					
		T2		CO2	4	Regression –Model Training and Evaluation	10%	10	15	0	0	25	1 (Offline) 1.25 (Online)
					5	Classification –Model Training and Evaluation	12%						
					6	Introduction to Deep Learning	3%						
		T2 Test Syllabus:					**25%**	**T2 Test Marks: 25**					
		T3		CO3	7	Web Scraping, APIs & Data Ingestion	13%	10	15	0	0	25	1 (Offline) 1.25 (Online)
					8	Django Framework (Backend-First Development)	12%						
		T3 Test Syllabus:					**25%**	**T3 Test Marks: 25**					
		Continuous and Comprehensive Evaluation (CCE) Total T1 to T3 Tests :					**75%**	**30**	**45**	**0**	**0**	**75**	
	Semester End Evaluation (SEE)	T4	T1	CO1	1.2.3	Data Analysis with Pandas & EDA Data Visualization with Python Introduction to Machine Learning with Python	25%	3	0	0	0	3	2 (Offline) 1.25 (Online)
			T2	CO2	4,5,6	Regression –Model Training and Evaluation Classification –Model Training and Evaluation Introduction to Deep Learning	25%	3	0	0	0	3	
			T3	CO3	7,8	Web Scraping, APIs & Data Ingestion Django Framework (Backend-First Development)	25%	3	0	0	0	3	
			T4	CO4	9	Django Models and Users	13%	16	25	0	0	41	
					10	REST APIs & Backend Services	12%						
		T4 Test Syllabus:					**100%**	**T4 Test Marks: 50**					
		Semester End Evaluation (SEE) Total T4 Test:					**25%**	**25**	**25**	**0**	**0**	**50**	